

ClimSim RNN

A vertical-sequence emulator for atmospheric sub-grid physics tendencies

The ClimSim Task



ClimSim is a hybrid ML-physics dataset from Columbia's LEAP Center. A high-resolution simulation is run alongside a coarse climate model; the gap between them is the “sub-grid” physics a coarse model misses.



The task: given one atmospheric column's state – 60 vertical levels of temperature, humidity, pressure, and surface fluxes – predict the tendencies a high-resolution run would produce for that column.

Atmospheric column

60 vertical levels × state (T, q, ps, fluxes)



Vertical-sequence RNN

reads the 60 levels as a sequence



Tendencies + surface fluxes

dT/dt , dq/dt (per level) + 8 surface variables

Framing the Problem

Vertical level as the sequence axis — levels are physically coupled by convection and vertical mixing, so that's the axis with an actual sequential relationship to model.



One sample = one column at one moment

60

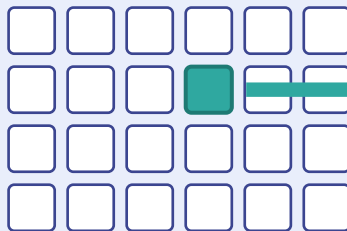
vertical levels form the sequence axis

~55K

(timestep, column) samples across 2 days

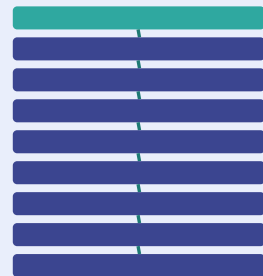
- Sequence axis: the 60 vertical levels, not calendar days or grid columns
- Each (timestep, grid column) pair is its own training sample
- Matches the framing ClimSim's own baseline emulator uses

grid columns × timesteps



one (timestep, column) pair

60 vertical levels



the sequence axis — coupled by convection & vertical mixing

Data Pipeline

Inputs & targets

Per-level: temperature and humidity (state_t, state_q0001). Aux scalars: surface pressure, insolation, latent/sensible heat flux.

Targets: dT/dt , dq/dt per level, plus 8 scalar surface fluxes (NETSW, FLWDS, PRECSC, PRECC, SOLS, SOLL, SOLSD, SOLLD).

Leak-free split

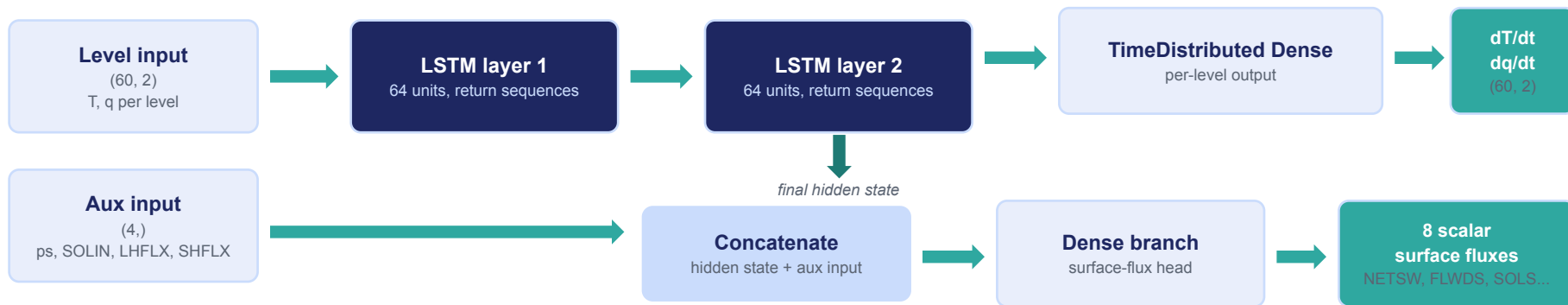
Day 1 split 90/10 into train/val (sklearn, fixed seed). Day 2 is held out entirely and used only as an unseen-day generalization test – never touched during training or normalization fitting.

Normalization, fixed at the root cause

Per-level mean/std fit on the train split only. An epsilon floor ($1e-8$) on std prevents divide-by-zero at near-constant levels (e.g. humidity at the top of the atmosphere) – replacing the legacy notebook's post-hoc NaN/Inf clipping with a fix at the source.

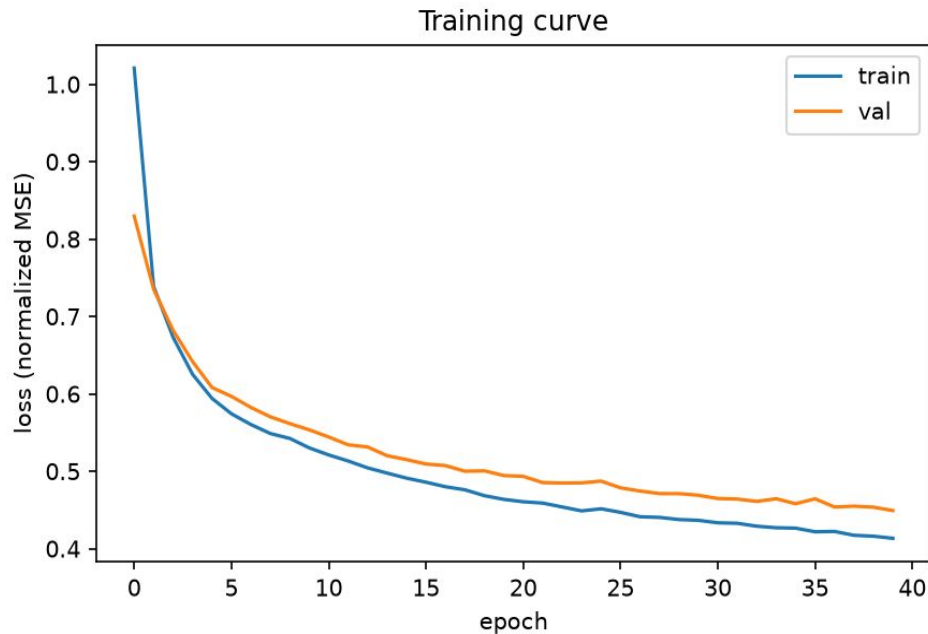
Model Architecture

A two-head design: level tendencies stay a per-level sequence output; scalar surface fluxes read off the final hidden state.



Physically-motivated split: per-level tendencies are inherently sequential, so they stay a TimeDistributed output along the level axis; surface fluxes are single column-level quantities, so they're read off the LSTM's summarized final state plus the auxiliary scalar inputs.

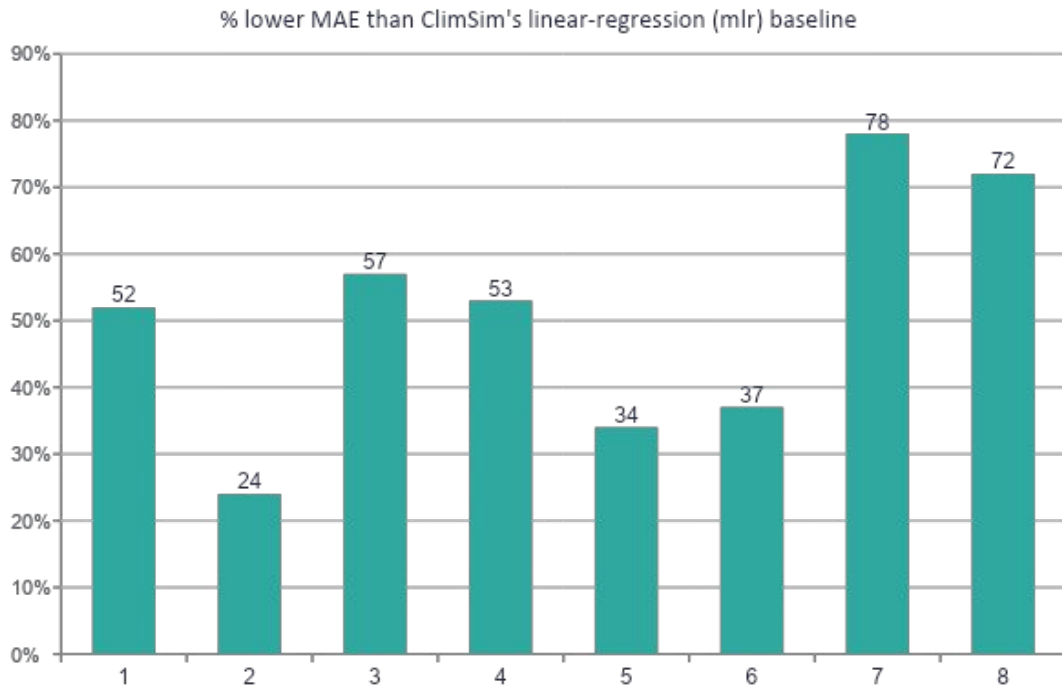
Training



Training setup

- Adam optimizer, MSE loss
- EarlyStopping on validation loss
- Trained on day 1 (train split); monitored on day 1 (val split)
- Evaluated only afterward, on day 2 – an entirely held-out day
- ~8 minutes end-to-end on Apple Silicon CPU (Python 3.12 / TensorFlow 2.21)
- 4/4 clean end-to-end re-runs with zero errors – verified reproducible

Results: Beats Every Baseline



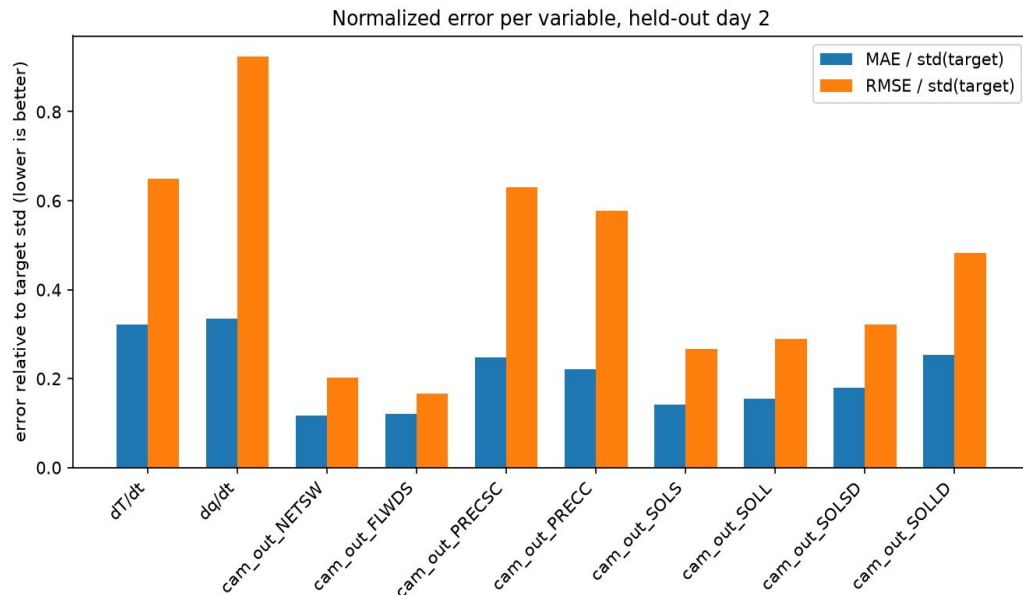
52–78%

lower MAE than ClimSim's tuned linear-regression baseline, across all 8 scalar surface variables

8 / 8

scalar variables beat both of ClimSim's own baselines (const and mlr), most by an order of magnitude

Error Across All Variables



Physical units span ~13 orders of magnitude here (K/s vs W/m²), so raw MAE/RMSE aren't comparable across variables. Normalizing by each variable's own standard deviation gives a scale-free "fraction of natural variability explained."

Tendency targets

- R^2 (dT/dt) = 0.58
- R^2 (dq/dt) = 0.15 – moisture is the harder target, consistent with ClimSim's own baselines
- Absolute MAE for these two isn't directly comparable to ClimSim's official numbers, which apply an energy-unit (pressure/area-weighted) conversion this repo doesn't reproduce – R^2 is the safer cross-check

Scalar surface fluxes

Same native units as ClimSim's archived baselines – directly comparable, and the source of the results on the previous slide.

Limitations



Two days of sample data

Demonstrates the approach; not tuned or competitive at the scale of ClimSim's full multi-year dataset.



Minimal feature set

Temperature, humidity, surface pressure, and three surface fluxes as input – not ClimSim's full input spec (winds, ozone, additional moisture species, etc.).



No energy-unit weighting

ClimSim's official leaderboard weights tendencies by pressure-level thickness and grid-cell area; this repo reports plain per-variable MAE/RMSE/ R^2 /bias in native units instead.



Untuned architecture

The 2-layer LSTM is a reasonable starting point for the vertical-sequence framing, not a tuned or benchmarked one.

Takeaways



Physically-motivated redesign

Sequence axis = the 60 vertical levels, not calendar days – matching how ClimSim's own baseline frames the problem.



Root-cause fixes, not patches

Leak-free normalization fit on train only; an epsilon floor replaces the legacy notebook's post-hoc NaN/Inf clipping.



Outperforms ClimSim's baselines

Beats both const and mlr baselines on all 8 comparable scalar variables, most by an order of magnitude.

Fully reproducible end-to-end in ~8 minutes – *README.md* + *notebooks/01_rnn_climate_tendency_emulator.ipynb*